Euclid MER VIS geometry outcomes summary

Completed local fitting and grouped galaxy review reports
Updated 20 October 2026

All 2777 galaxy cutouts now have an individual PNG report and a place in the grouped PDF collection. The fitting returned 2207 provisional geometry estimates and 570 failures. We have separated reports with detected quality issues so they can be reviewed efficiently before adopting geometry for deprojection.

Fitting outcomes

Result	Galaxies	Share of sample
Fit with no automatic geometry flags	546	19.7%
Fit with one or more geometry flags	1661	59.8%
Fit failed	570	20.5%
Total	2777	100.0%

A returned fit supplies observed disc PA, observed bar PA and inclination. It remains provisional until the selected light contours have been checked against the image. A failed fit remains visible in the reports, with unavailable geometry rather than zero-valued angles.

The review groups

Group	PNG reports	PDF batches
QI quality issues	2250	38
NoQI no detected issue	527	9
Total	2777	47

QI includes any geometry flag, a failed fit or partial coverage of the requested cutout. It contains 1661 flagged fits, 570 failures and 19 otherwise unflagged fits with partial coverage. These three categories are disjoint and total 2250. NoQI contains the remaining 527 galaxies. NoQI is a screening label, not scientific approval.

The 47 PDFs contain 463 pages at six galaxies per page, with at most 60 galaxies per batch. QI and NoQI reports never share a PDF. Each PNG is 2000 × 1750 pixels. The original FITS images and source fitting manifest remain unchanged.

Scope and basis

The sample comprises 2777 Euclid Q1 candidate barred spirals selected from Galaxy Zoo morphology, with cutouts drawn from 309 mosaics. This summary uses the completed fitting run and the verified report collection. Report generation reused the existing ellipse profiles; it did not rerun the scientific fits.

Quality issues and fitting failures

Most detected issues arise from a small selected bar interval, unstable disc orientation or changing outer ellipticity. Flags describe specific numerical conditions. They may overlap, so their counts must not be added to infer a number of affected galaxies.

Quality condition	Count	Share of 2207 fits
Bar interval upper bound below 1.5 arcsec	1210	54.8%
Disc PA radial scatter above 10°	791	35.8%
Disc ellipticity span above 0.15	573	26.0%
Nearly face on $q > 0.90$	200	9.1%
Bar PA radial scatter above 10°	28	1.3%
Median bar ellipticity below 0.25	7	0.3%

No high-inclination or axis-ratio-below-thickness flags occurred. A small-bar warning indicates sensitivity to resolution and interval selection; its interval boundary is not a measured physical bar length. Disc PA instability warns that the adopted outer contours do not share a consistent direction. Disc ellipticity instability makes the inclination estimate particularly sensitive to the chosen range.

Why 570 galaxies failed

Recorded reason	Galaxies
Too few reliable samples in the selected bar or disc interval	431
Fewer than ten reliable isophotes overall	138
Catalogue centre not on a detected component	1

The 431 interval-selection failures are the largest potential recovery group. A reviewed interval may help when enough suitable light is available, but merely widening an interval until the calculation succeeds does not establish a physical bar or disc. Short profiles may need a better mask, centre, background estimate or a different fitting approach.

Coverage needs a separate check

There are 62 partial requested cutouts, including 51 fitted objects and 19 unflagged fits. The requested box crosses a mosaic boundary; this does not necessarily mean the galaxy is truncated. All 62 reports enter QI, including those already flagged or failed. The 19 unflagged partial cases explain the difference between 546 unflagged fits and 527 NoQI reports.

The unflagged fits have a median inclination of 53.2°, compared with 43.8° for all fits. This reflects geometry-dependent measurement difficulty and the face-on warning threshold. The screened subset should not be treated as an unbiased sample of the underlying barred galaxy population.

How Codex calculates position angle

Codex fits ellipses to contours of constant brightness using Photutils. Photutils initially measures the ellipse major axis from the positive image x-axis towards the positive image y-axis, counterclockwise in a plot where x increases rightwards and y upwards. This intermediate pixel angle depends on the FITS orientation.

The code then uses the FITS World Coordinate System (WCS) at the galaxy centre to convert that pixel direction into local celestial East and North components. In sky_pa(), the calculation is PA = degrees(atan2(East, North)) modulo 180°. The local matrix stores West and North, so the code passes minus West as the East component. This accounts for image rotation and reflection.

The reported sky convention

Both fitted disc PA and fitted bar PA are measured FROM CELESTIAL NORTH TOWARDS EAST. North–South is 0°, northeast–southwest is 45°, East–West is 90°, and southeast–northwest is 135°. The range is 0° inclusive to 180° exclusive. An ellipse axis has no arrow, so 0° and 180° describe the same direction.

On a North-up, East-left astronomical display this direction is COUNTERCLOCKWISE. On a North-up, East-right display it is CLOCKWISE. Thus North towards East is the unambiguous definition; clockwise or counterclockwise alone depends on the displayed image orientation. A mirrored display does not change the WCS-based catalogue PA.

The code calculates an axial mean over reliable isophotes in the selected outer disc interval for disc PA and the selected inner bar interval for bar PA. It doubles angles before averaging and halves the result, so axes near 179° and 1° average near 0°, rather than 90°. These are observed sky angles before deprojection. Inclination is a separate quantity calculated from the disc axis ratio.

How to interpret the 90 degree adjustment

A constant 90° offset can represent perpendicular reference axes with the same measurement sense, or a major-axis versus minor-axis choice. It does not by itself establish which explanation applies. In particular, changing from counterclockwise from East to the astronomical North-towards-East convention on a North-up, East-left display also involves a change of measurement sense; one cannot assume that it always means simply adding 90°.

Euclid documents raw POSITION_ANGLE as counterclockwise from the image x-axis, SExtractor THETA_IMAGE, in the range −90° to +90°. The image x-axis is not necessarily celestial East. Our project first converts that pixel direction through WCS, then applies the adopted +90° adjustment once. We use this as the adopted perpendicular-axis correction for this Q1 comparison, without claiming that Euclid officially measures its raw angle from celestial East or North.

Code references: fit_MER_VIS_geometry_portable.py, sky_pa() and axial_mean(); MER_VIS_geometry_transforms.py, local_pixel_matrix(). Euclid definition: https://euclid.esac.esa.int/dr/q1/dpdd/merdpd/dpcards/mer_finalcatalog.html#mer-main-catalog-fits

Comparison with Euclid geometry

Adopted position angle adjustment

The adopted project conversion is adjusted Euclid sky PA = (previously converted Euclid sky PA + 90°) modulo 180°. The reports use this adjusted value consistently. It changes the catalogue comparison direction; it does not change the raw catalogue measurement, rotate the FITS image or alter the independently fitted Photutils angles. The adjustment must be applied once only.

Comparison with Photutils	All 2207 fits	546 unflagged fits
Adjusted detection PA median absolute difference	7.07°	3.72°
Adjusted detection PA within 10°	59.4%	79.3%
Detection inclination proxy median absolute difference	4.28°	2.71°
Single Sérsic inclination proxy median absolute difference	8.41°	6.76°

The 546-object comparison subset is defined by absence of geometry flags and includes the 19 partial cases. It is not the same subset as the 527 NoQI reports. The completed validation of 51 stable non-diagonal discs supports the adopted PA adjustment, with a median absolute difference of 2.15°. This summary does not repeat the convention analysis.

What the agreement establishes

Euclid detection and single Sérsic shapes describe whole-object light. Our disc geometry comes from a selected outer interval. Differences can therefore be physical or methodological, particularly in galaxies with bulges, bars, rings, arms or neighbours. Agreement supports consistency but does not prove that either measurement gives the true outer-disc geometry.

No independent Euclid bar PA is available for this sample, and separate disc-component axis ratios are unpopulated. The reports explicitly mark the missing bar counterpart. Euclid inclination values are proxies derived from axis ratios under the same circular thin-disc assumption, $q_0 = 0$, used in the local fit.

Angle and delta conventions

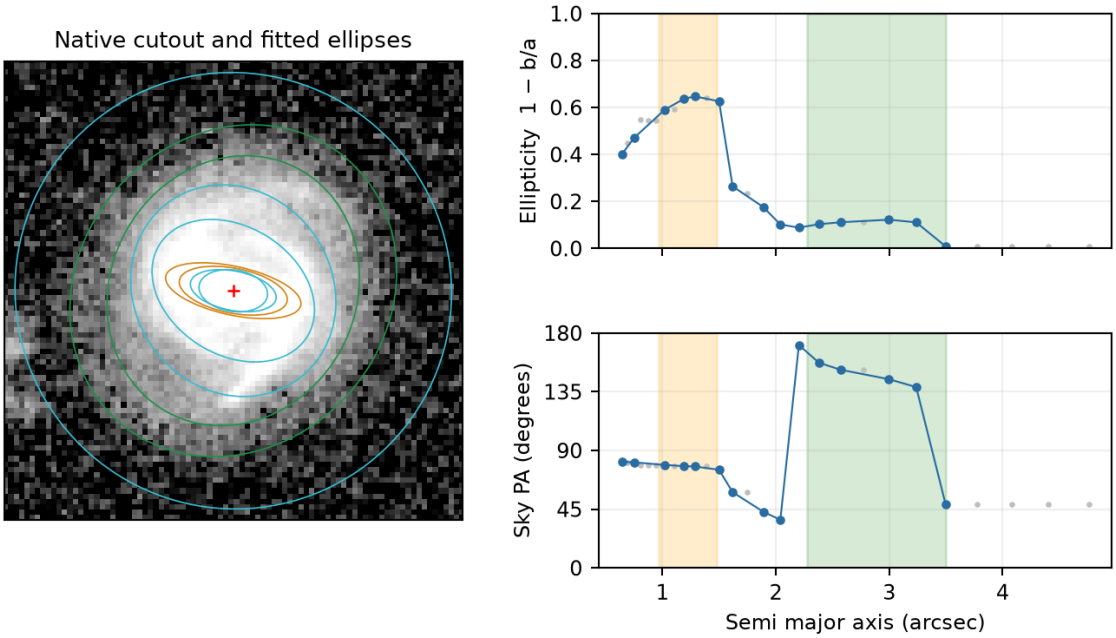
All reported angles use degrees. Sky PA is east of north modulo 180°; inclination is 0° face on and 90° edge on. Deltas are Photutils minus Euclid. PA differences use the shortest signed axial separation in [−90°, 90°), while inclination differences use ordinary subtraction. Display values round to one decimal place after calculating the deltas.

Bar classifications remain separate

The sample selection uses Galaxy Zoo Euclid Q1 morphology from Zenodo record 15106473. Strong and weak bar fractions describe predicted classifications, not a physical torque-based bar strength or a bar PA. The fitting and QI labels should therefore not be interpreted as confirming or rejecting every catalogue bar classification.

Review priorities and use of the reports

Begin with the QI PDFs and use the image and both profiles together. Check whether the centre lies on the nucleus, whether nearby light affects the ellipses, and whether the selected outer interval follows a stable disc. Nearly circular contours can have an ill-defined PA even when the numerical fit completes.



Illustrative QI case −515488559271594427. The outer PA changes markedly across the selected green disc interval. Orange indicates the selected bar interval. A returned mean angle or close Euclid agreement cannot establish stable outer-disc geometry in this case.

Review group	Recommended action
431 interval-selection failures	Inspect profiles and choose defensible bar and disc ranges where the image supports them.
138 short reliable profiles and one centre failure	Inspect signal, masks, catalogue centre and neighbouring light before choosing a recovery method.
Flagged fits and partial cutouts	Review the specific warnings. Keep a usable disc measurement separate from an uncertain bar measurement.
527 NoQI reports	Check the physical component identification before adopting geometry.

Record reviewed centres, masks, radial ranges and accepted values in an adopted-geometry table before generating final deprojected, bar-aligned images. Preserve failed measurements and the reasons for exclusion. A geometry flag is a reason for review rather than an automatic rejection of every quantity.

Finding individual galaxies

Reports are under Euclid/MER_VIS_galaxy_reports/all_2777 in the research data folder. The png_QI and pdf_QI folders contain quality-issue reports, and png_NoQI and pdf_NoQI contain the other reports. MER_VIS_report_review_index.csv identifies each galaxy's PNG, PDF batch, page and position. The associated metadata retains the exact original flag strings and full-precision values.

Sources: MER_VIS_local_geometry_manifest.csv; MER_VIS_report_summary.json; MER_VIS_report_verification.json; MER_VIS_report_review_index.csv; the completed PA comparison summaries; and the revised Euclid MER VIS galaxy geometry fitting report. Counts and comparisons refer to the completed October 2026 local run.